

Tier 0 structural/ control

n = 40 judged pairs per family

syntactic_scramble

span 1 of 2

word order in MOVE/SUBTASK

move to the

→ ∅

judge: meaning preserved

1.000

cross_task_swap

span 1 of 10

whole trace ← another task

∅

→ because it's the closest and most direct path

judge: meaning preserved

0.025

Tier 1 word-level semantic

n = 40 judged pairs per family

direction_flip

span 1 of 2

left/right, up/down in MOVE

left

→ **right**

judge: meaning preserved

0.200

gripper_flip

span 1 of 2

grasp/release in PLAN

grasp

→ **release**

judge: meaning preserved

0.050

verb_swap

span 1 of 3

action verb → unrelated verb

move

→ **hold**

judge: meaning preserved

0.575

negation

span 1 of 2

prepend to SUBTASK/MOVE

∅

→ **do not**

judge: meaning preserved

0.100

Tier 2 object / causal chain

n = 24–29 judged pairs per family

subject_swap

span 1 of 5

referent → visible distractor

flat stove.

→ **white bowl.**

judge: meaning preserved

0.000

location_swap

spatial phrase in PLAN/TASK

left

→ **right**

judge: meaning preserved

0.276

adversarial_plausible

span 1 of 5

referent → 2nd-most-visible

flat stove.

→ **wooden two layer shelf.**

judge: meaning preserved

0.000

Tier N calibration nulls

n = 40 judged pairs per family

paraphrase_null

span 1 of 6

synonym in MOVE/SUBTASK

move

→ **shift**

judge: meaning preserved

0.975

bbox_jitter_null

span 1 of 3

±1px on VISIBLE OBJECTS

[[106, 165], [146, 207]],

→ **[[105, 166], [147, 206]],**

judge: meaning preserved

1.000

identity_control

nothing ($X \rightarrow X$)

(byte-identical)

judge: meaning preserved

1.000